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23DEA3202: DATA EXPLORATION AND PREPARATION

Semester	:	**B. Tech (VI semester)**
Name of the Student	:	**PHANI TEJASWI.P**
Roll Number	:	**2320030069**
Major Project Title	:	**Traffic Congestion Patterns in Smart Cities**
Domain Name	:	**Smart Cities & IoT**
Base Paper Title	:	**Traffic Congestion Pattern Analysis Using Smart City Data**

Abstract:
Urban traffic congestion is one of the most significant challenges faced by modern smart cities, as it adversely affects mobility, economic productivity, environmental sustainability, and overall quality of life. Rapid urbanization, population growth, and the increasing number of vehicles have intensified the need for efficient and data-driven traffic management systems. The widespread adoption of Internet of Things (IoT) technologies—such as traffic sensors, GPS-enabled vehicles, and surveillance systems—has enabled the continuous generation of large volumes of traffic-related data in smart city environments.
This project focuses on **Data Exploration and Analysis (DEA)** of urban traffic congestion data to understand spatio-temporal congestion patterns. The analysis investigates variations in traffic volume and congestion levels across different time periods, including peak and off-peak hours, as well as comparisons between weekdays and weekends. In addition, location-based analysis is performed to identify congestion-prone zones within the city. Descriptive statistical techniques and data visualization methods are employed to explore trends, recurring patterns, and anomalies present in the traffic data.
The findings of this exploratory study reveal consistent congestion trends, critical time intervals with high traffic density, and urban areas that regularly experience traffic bottlenecks. These insights can assist urban planners and traffic authorities in understanding traffic behavior and support data-driven decision-making for improved traffic management and sustainable smart city development.

Keywords: Traffic Congestion, Smart Cities, Data Exploration and Analysis, Urban Traffic

Recommended / Not Recommended (NR)
If NR, Remarks: _____

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